15-442/15-642: Machine Learning Systems

Transformer and Attention

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Outline

Sequence Prediction

Transformer and Self-Attention

Recursive Attention

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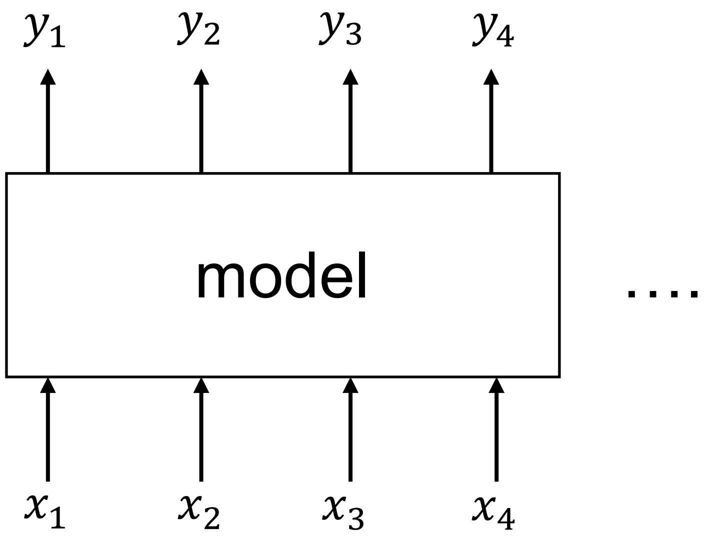
Sequence Prediction

Take a set of input sequence, predict the output sequence

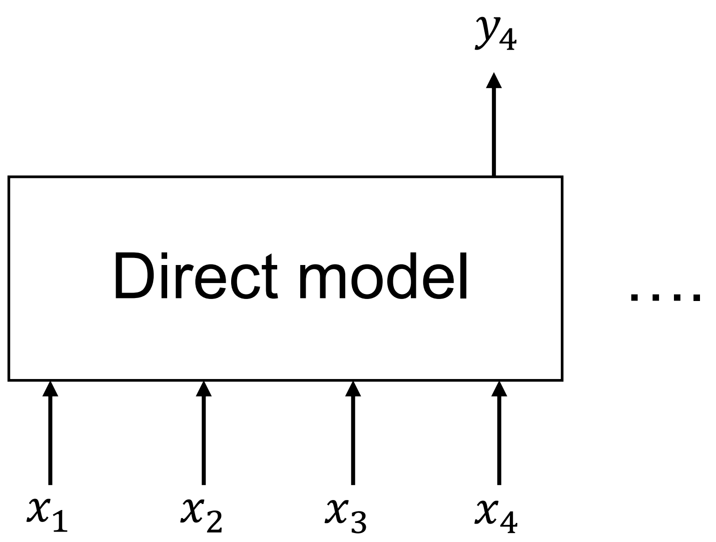


Predict each output based on history $\quad y_t = f_\theta(x_{1:t})$

There are many ways to build up the predictive model

“Direct Prediction”

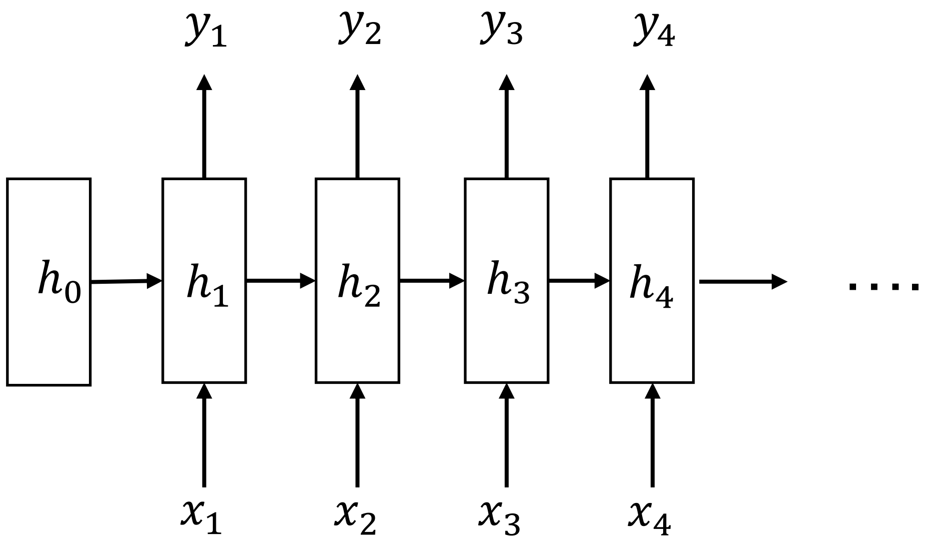
One approach is we can do “direct prediction”



Challenge: the function needs to make prediction based on different size inputs.

RNN Approach

Try to maintain a "latent state" that is derived from history



The information is carried only through h_t

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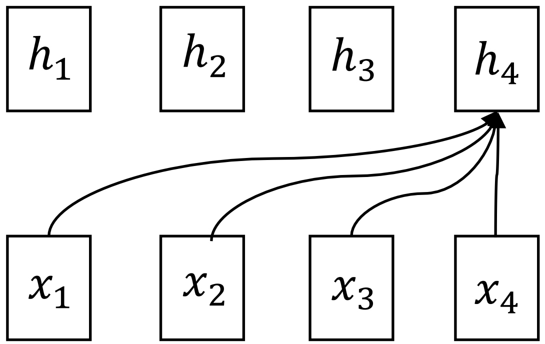
“Attention” Mechanism

Generally refers to the approach that weighted combine individual states

Attention output

Hidden states from previous layer



$$h_t = \sum_{i=1}^{t} s_i x_t$$

Intuitively s_i is “attention score” that computes how relevant the position i’s input is to this current hidden output

There are different methods to decide how attention score is being computed

Self-Attention Operation

Self attention refers to a particular form of attention mechanism.

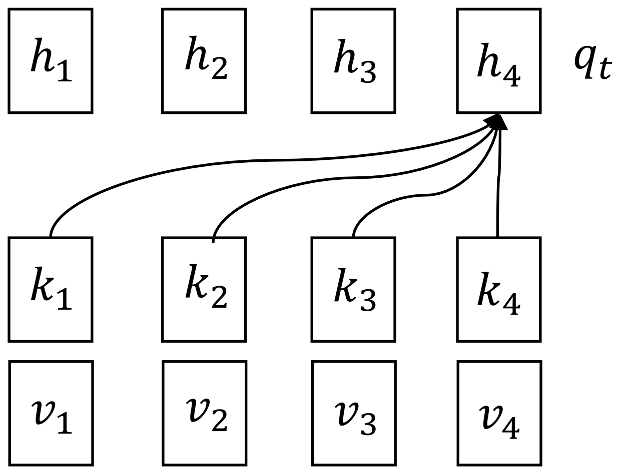
Given three inputs $Q, K, V \in \mathbb{R}^{T \times d}$ ("queries", "keys", "values")

Define the self-attention as:

$$\mathrm{SelfAttention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{d^{1/2}}\right)V$$

A Closer Look at Self-Attention

Use q_t, k_t, v_t to refers to row t of the K matrix



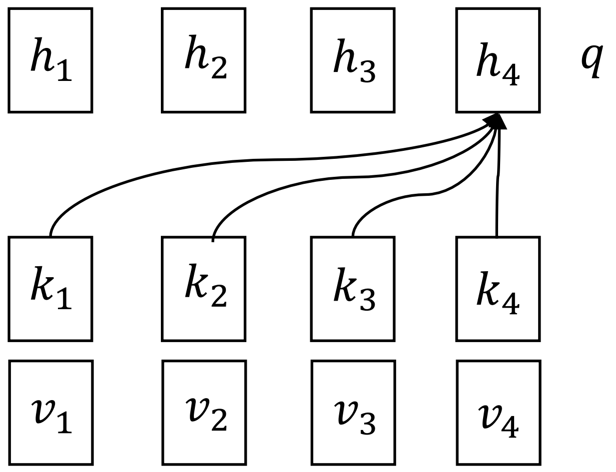
Ask the following question:

How to compute the output h_t, based on q_t, K, V one timestep t

To keep presentation simple, we will drop suffix t and just use q to refer to q_t in next few slide

A Closer Look at Self-Attention

Use q_t, k_t, v_t to refers to row t of the K matrix



Conceptually, we compute the output in the following two steps:

Pre-softmax “attention score”

$$s_i = \frac{1}{\sqrt{d}} q k_i^T$$

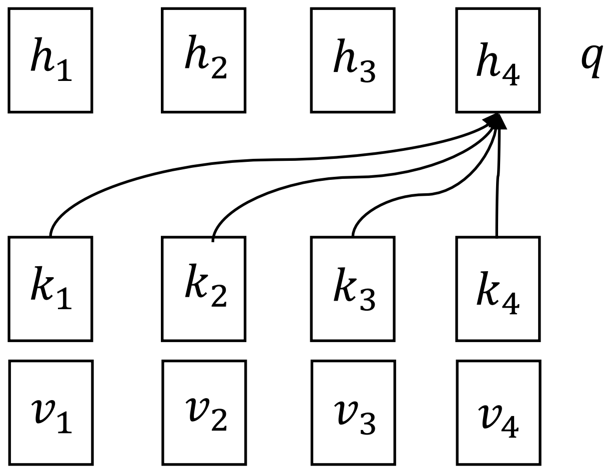
Weighed average via softmax

$$h = \sum_i \mathrm{softmax}(s)_i v_i = \frac{\sum_i \exp(s_i) v_i}{\sum_j \exp(s_j)}$$

Intuition: s_i computes the relevance of k_i to the query q,
then we do weighted sum of values proportional to their relevance

Comparing the Matrix Form and the Decomposed Form

Use q_t, k_t, v_t to refers to row t of the K matrix



$$\text{SelfAttention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{d^{1/2}}\right)V$$

Pre-softmax "attention score"

$$S_{ti} = \frac{1}{\sqrt{d}} q_t k_i^T$$

Weighed average via softmax

$$h_t = \sum_i \text{softmax}\left(S_{t,:}\right)_i v_i = \text{softmax}\left(S_{t,:}\right)V$$

Intuition: s_i computes the relevance of k_i to the query q,
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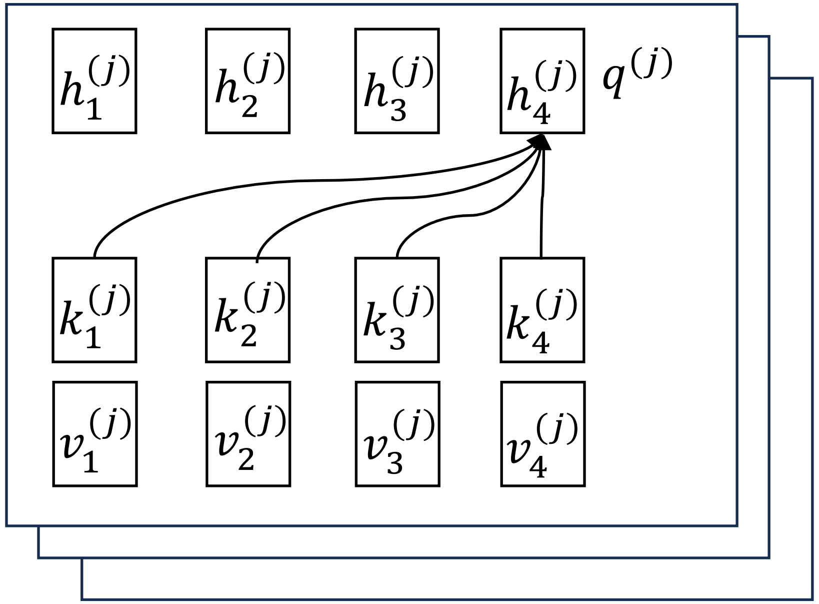
Multi-Head Attention

Have multiple "attention heads" $Q^{(j)}, K^{(j)}, V^{(j)}$ denotes j-th attention head



Apply self-attention in each attention head

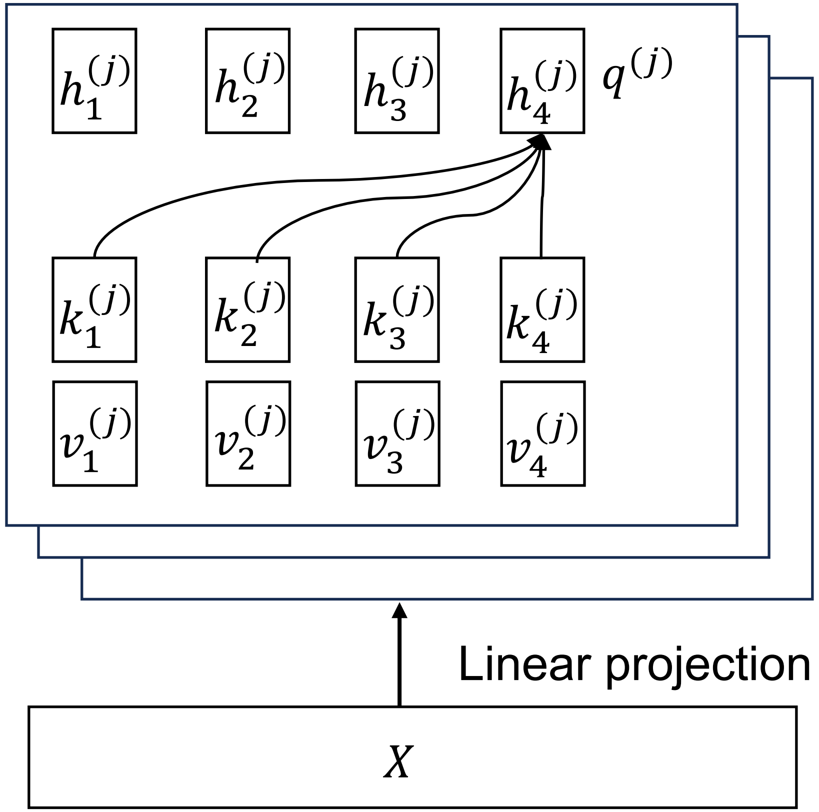
$$\mathrm{SelfAttention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{d^{1/2}}\right)V$$

Concatenate all output heads together as output

Each head can correspond to different kind of information.
Sometimes we can share the heads: GQA(group query attention) all heads share K, V but have different Q

How to get Q K V?

Obtain Q, K, V from previous layer's hidden state X by linear projection



$$Q = XW_q$$
$$K = XW_K$$
$$V = XW_V$$

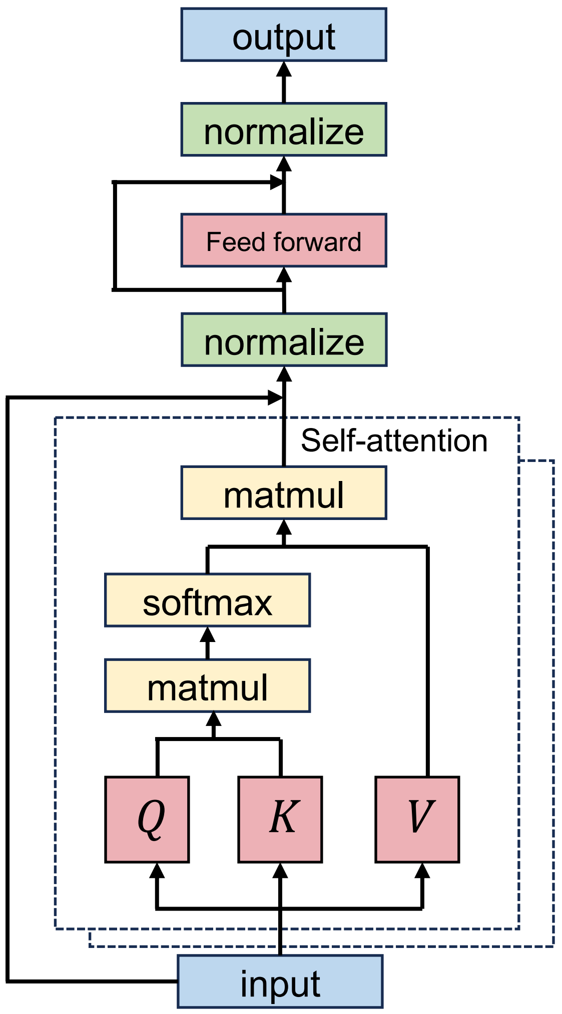
Can compute all heads and Q, K, V together then split/reshape out into individual Q, K, V with multiple heads

Transformer Block

A typical transformer block

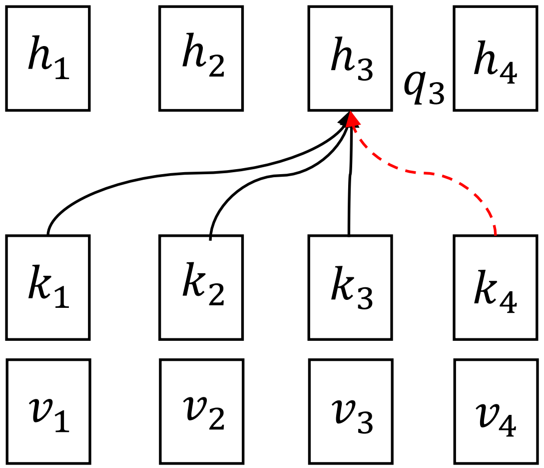
$$Z = \mathrm{SelfAttention}\left(XW_K, XW_Q, XW_V\right)$$
$$Z = \mathrm{LayerNorm}(X + Z)$$
$$H = \mathrm{LayerNorm}(\mathrm{ReLU}(ZW_1)W_2 + Z)$$

(multi-head) self-attention, followed by a linear layer and ReLU and some additional residual connections and normalization



Masked Self-Attention

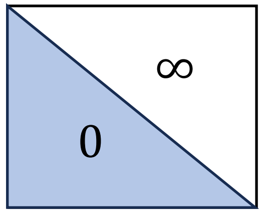
In the matrix form, we are computing weighted average over all inputs



In auto regressive models, usually it is good to maintain casual relation, and only attend to some of the inputs (e.g. skip the red dashed edge on the left). We can add “attention mask”

$$\mathrm{MaskedSelfAttention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{d^{1/2}} - M\right)V$$

$$M_{ij} = \begin{cases} \infty, j > i \\ 0, j \leq i \end{cases}$$



Only attend to previous inputs. Depending on input structure and model, attention mask can change.

We can also simply skip the computation that are masked out if there is a special implementation to do so

Discussions

What are the advantages of transformers versus RNNs

What are the disadvantages

What are other possible ways to apply attention mask

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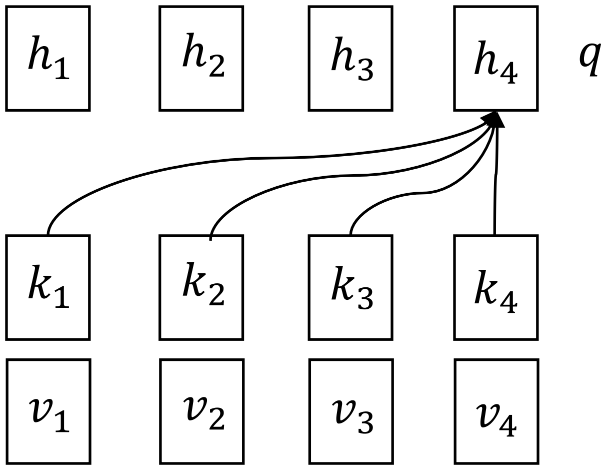
Self-Attention Recap

Use q_t, k_t, v_t to refers to row t of the K matrix



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Generalizing Attention Score and Value Vector

Pre-softmax "attention score" $s_i = \frac{1}{\sqrt{d}} q k_i^T$

Define the following "Attention weight" for an index set I

$$\mathrm{s}(I) = \log\left(\sum_{i \in I} \exp(s_i)\right)$$

Let us generalize the value vector v for index set I

$$\mathrm{v}(I) = \sum_{i \in I} \mathrm{softmax}(s)_i v_i = \frac{\sum_{i \in I} \exp(s_i) v_i}{\exp(s(I))}$$

Reference: flashinfer.ai

Generalizing Attention Score and Value Vector

Pre-softmax "attention score" $s_i = \frac{1}{\sqrt{d}} q k_i^T$

$$\mathrm{s}(I) = \log(\sum_{i \in I} \exp(s_i)) \text{ , } \mathrm{v}(I) = \sum_{i \in I} \mathrm{softmax}(s)_i v_i = \frac{\sum_{i \in I} \exp(s_i) v_i}{\exp(s(I))}$$

When index set $I = \{i\}$, $\mathrm{s}(\{i\}) = s_i$, $\mathrm{v}(\{i\}) = \mathrm{v}_i$

When index set $I = \{1,2, \ldots t\}$, $\mathrm{v}(I)$ is the final output of the attention

Recursive Attention

$$\mathrm{s}(I) = \log(\textstyle\sum_{i \in I} \exp(s_i)) \text{ , } \mathrm{v}(I) = \sum_{i \in I} \mathrm{softmax}(s)_i v_i = \frac{\sum_{i \in I} \exp(s_i) v_i}{\exp(s(I))}$$

For any partition $\{I_j\}$ of I such that $I = \bigcup_{j=1}^{n} I_j$, the following relation holds

$$\mathrm{s}(\textstyle\bigcup_{j=1}^{n} I_j) = \log(\sum_j \exp(s(I_j))), \ \mathrm{v}(\bigcup_{j=1}^{n} I_j) = \sum_j \mathrm{softmax}([s(I_1), s(I_2) \ldots])_j v(I_j)$$

We can use the same rule to recursively combine the vector and “attention score” of any subsets of indices

When we obtain the value vector of all indices, that becomes the attention output

Discussions: Recursive Attention

$$\mathrm{s}(I) = \log\left(\sum_{i \in I} \exp(s_i)\right), \ \mathrm{v}(I) = \sum_{i \in I} \mathrm{softmax}(s)_i v_i = \frac{\sum_{i \in I} \exp(s_i) v_i}{\exp(s(I))}$$

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Attention computation is **communicative** and **associative** and can be done in a divide-and-conquer fashion. An important property for a lot of system optimizations

Discussion: what can we do with this property?